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The Gender *Pay Gap*

Research Question & Hypotheses

- RQ: Does the effect of gender on salary differ depending on employees' educational attainment and age?
    - H1: Among employees with advanced degrees, the gender gap in salary is smaller than among those with Bachelor's degrees.
    - H2: Among older employees with advanced degrees, the gender gap salary is larger than younger employees with advanced degrees.

Method

Variables

IVs:

- *sex_2023* for gender (M or F)
- *Age*
- *d2dg* - education attainment

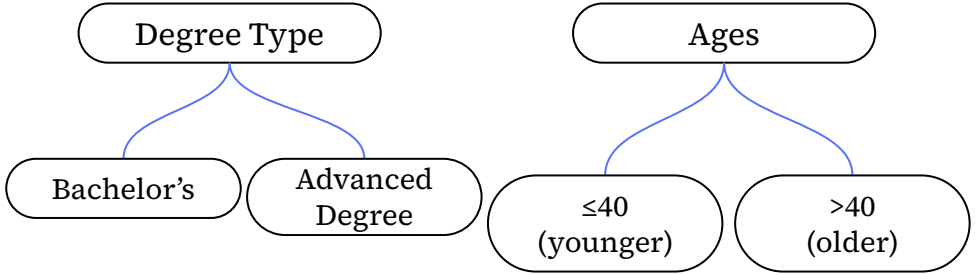
DV:

- *Salary* (USD)

Population of Interest

- U.S. full-time workers (more than 40 hours a week)
- Located in the U.S. and are U.S. citizens
- Controlled for having kids

Collapsed Variables



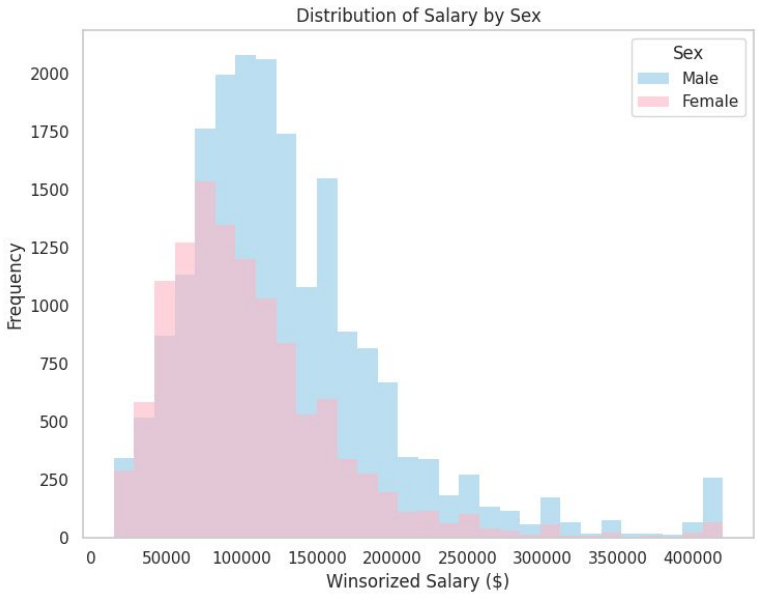
A few important considerations

- Excluding part-time workers limits generalizability, as women often work part-time due to caregiving.
- Focusing on U.S. workers reduces applicability to international contexts.
- Controlling for children doesn't fully capture the impact of parenthood on women's salaries.

See Hex Notebook for more detail

Basic Stats on the Gender Gap

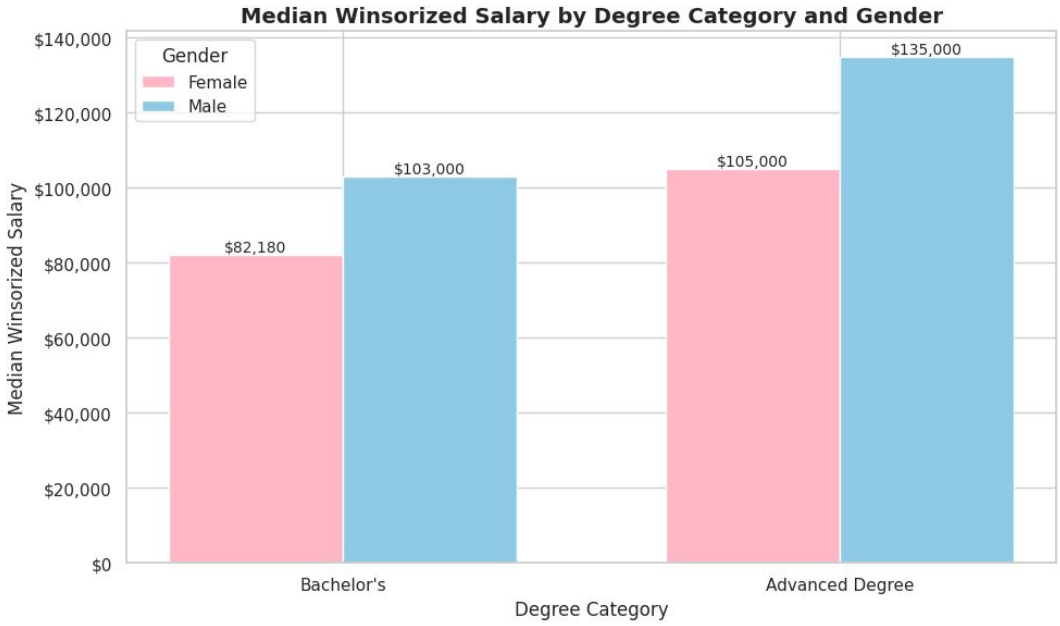
Men earn higher salaries than women, with both higher mean and median salaries.



The salary distribution for men is more spread out, with more individuals in higher salary ranges, while women's salaries are more concentrated in the middle and lower ranges.

HYPOTHESIS 1

H1: Among employees with advanced degrees, the gender gap in salary is **smaller** than among those with Bachelor's degrees.



Gender salary gaps:

- Salary gap for Bachelor's employees is **$20820** ($103000-$82180).
- Salary gap for Advanced degrees' employees is **$30000** ($135000-$105000).
- Gender gap in salary is **larger** for employees with Advanced degrees than those with Bachelor's degrees.
    - This is the **opposite** of our hypothesis 1 -> **does not support H1**

Hypothesis 1: Regression Results

```
Gender × Degree Category Model:
                            OLS Regression Results
==============================================================================
Dep. Variable:             log_salary   R-squared:                       0.113
Model:                            OLS   Adj. R-squared:                  0.113
Method:                 Least Squares   F-statistic:                     1006.
Date:                Thu, 11 Dec 2025   Prob (F-statistic):               0.00
Time:                        04:29:47   Log-Likelihood:                -24596.
No. Observations:                31544   AIC:                         4.920e+04
Df Residuals:                    31539   BIC:                         4.924e+04
Df Model:                            4
Covariance Type:            nonrobust
==================================================================================================
                                             coef    std err          t      P>|t|      [0.025      0.975]
--------------------------------------------------------------------------------------------------
Intercept                                 11.2333      0.007   1570.114      0.000      11.219      11.247
C(degree_category)[T.Advanced Degree]      0.2453      0.010     25.279      0.000       0.226       0.264
sex                                        0.2160      0.008     25.567      0.000       0.199       0.233
sex:C(degree_category)[T.Advanced Degree]  0.0091      0.012      0.741      0.459      -0.015       0.033
chlvin                                     0.1730      0.006     28.705      0.000       0.161       0.185
==============================================================================
Omnibus:                     2436.381   Durbin-Watson:                   1.905
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             4799.014
Skew:                          -0.534   Prob(JB):                         0.00
Kurtosis:                       4.585   Cond. No.                         8.10
==============================================================================
```

```
Men with Bachelor's earn 24.11% more than women.
Men with Advanced Degrees earn 25.25% more than women.
Difference in gender gaps (Advanced - Bachelor): 1.14 percentage points.
```

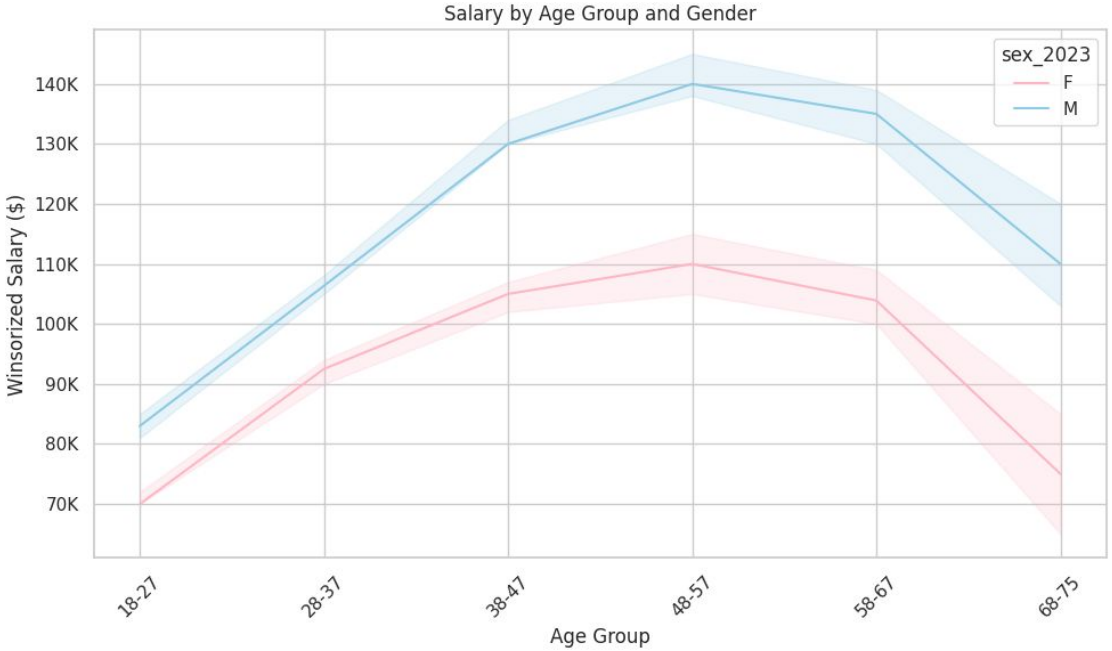
Regression results also does not support H1

- The interaction effect (Sex × Advanced Degree) -> The gender wage gap is about 1.14 percentage points larger among Advanced degree holders than among Bachelor's degree holders.

- This difference is **not statistically significant** ($p = 0.459$)
  - This observed difference is not statistically significant-> **does not support H1**.

HYPOTHESIS 2

H2: Among older employees with advanced degrees, the gender gap salary is **larger** than younger employees with advanced degrees.



Data Overview:

- Salaries increase with age, peaking at 48–57, then decline.
- Men earn more than women at all ages.
- The gender pay gap is largest in mid-career.
- Both genders see a noticeable drop in late career due to retirement planning.

A closer look… separating older and younger workers

Figure 1

Figure 2: Dark blue trendline indicates an increase in gap as age increases.

Younger group: **Men** average $112,830 vs. **women** $95,923 (gap: $16,907, 17.6%)
Older group: **Men** average $148,083 vs. **women** $120,933 (gap: $27,150, 22.5%)
The absolute wage gap increases by $10,243 between age groups.
This can also be seen on a more granular level in *figure 2*.

Regression Output for H2

The regression results strongly support that older employees with advanced degrees face a significantly larger gender wage gap than younger counterparts

```
Degree Category and Age Group Model:
                            OLS Regression Results
==============================================================================
Dep. Variable:              log_salary   R-squared:                       0.135
Model:                             OLS   Adj. R-squared:                  0.135
Method:                  Least Squares   F-statistic:                     985.3
Date:                 Thu, 11 Dec 2025   Prob (F-statistic):               0.00
Time:                         04:29:47   Log-Likelihood:                -24200.
No. Observations:                31544   AIC:                         4.841e+04
Df Residuals:                    31538   BIC:                         4.846e+04
Df Model:                            5
Covariance Type:            nonrobust
==========================================================================================
                                     coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
Intercept                         11.4295      0.007   1581.681      0.000      11.415      11.444
sex_2023[T.M]                      0.1917      0.008     23.603      0.000       0.176       0.208
C(degree_category)[T.Bachelor's]  -0.2413      0.006    -40.625      0.000      -0.253      -0.230
age_group                          0.1490      0.010     15.146      0.000       0.130       0.168
sex_2023[T.M]:age_group            0.0334      0.012      2.721      0.007       0.009       0.057
chlvin                             0.1406      0.006     23.191      0.000       0.129       0.152
==============================================================================
Omnibus:                     3267.613   Durbin-Watson:                   1.907
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             6956.073
Skew:                          -0.656   Prob(JB):                         0.00
Kurtosis:                       4.889   Cond. No.                         8.67
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
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- Growing Gender Gap: The positive interaction coefficient ($\beta = 0.0334$) indicates that the male wage advantage over females increases by approximately 3.3% with each one-unit increase in age group, holding other factors constant.
- For each additional age group: +3.3% wider gap
- Statistical Significance: $p = 0.007$ ($p < 0.01$)

After controlling for education level and family status, older employees with advanced degrees face a significantly larger gender wage gap than their younger counterparts. This supports the hypothesis and suggests that **career-stage penalties accumulate over time for highly educated women**.

Conclusion

Hypothesis 1

Not supported-> Among employees with advanced degrees, the gender gap in salary is not smaller than those with Bachelor's degrees.

Hypothesis 2

Supported -> Older employees with advanced degrees face a significantly larger gender wage gap than younger counterparts.

Implications

1. **Higher education alone does not close gender pay gaps**—advanced degrees raise salary but do not reduce disparities.
2. **Gender pay gaps widen with age among higher degree workers**—suggesting cumulative disadvantages over the career lifecycle.
3. **Organizational policies play a important role in pay equity**—greater transparency, standardized evaluations, and flexible career paths are critical to reducing long-term gaps.

Limitations

1. **Winsorized salaries may understate true gender pay gaps**—while winsorization reduces the influence of extreme values, it can also understate disparities at the top of the earnings distribution, where gender gaps are often the largest.
2. **Unobserved differences in hours worked may affect salary comparisons**—without controlling for worked hours, observed gaps may reflect worked hour differences rather than education or age effects.

See Hex Notebook for more detail.